

Assignment 2: End-to-End ML Versioning

Evidence-based final report - Git, DVC, Google Drive and Fashion-MNIST ANN

Student	Zainab Fatima	Repository	github.com/Aenonic/fashion-ann-pipeline
Dataset	Fashion-MNIST (70,000 images)	Final model	Dense 256, 10 epochs
Final result	Accuracy 87.81% / loss 0.3474	Audit date	4 October 2026

Part A - Git Fundamentals and Advanced Commands

The repository has an unsquashed history. The reflog independently preserves the destructive-history exercises (soft/hard reset) and the completed rebase; the merge commit preserves the Part E conflict resolution.

Requirement	Verified evidence	Meaning
A1-A2	Initial commit d21ef1b on main; dev branch created; 9 incremental Part B-D commits before v2.	Branch isolation and incremental history are present.
A3 log variants	--oneline --graph --all : topology; --stat -3 : recent file counts; -p -1 : exact patch; main..dev : commits reachable only from dev.	Each exposes structure or change detail omitted by plain log.
A4 diff variants	git diff : unstaged; --staged : index; main..dev : tip-to-tip; main...dev : merge-base-to-dev.	Two-dot includes both tips; three-dot isolates the dev-side work since divergence.
A5 stash	Workflow documented: stash push -u, stash list, branch switch, stash pop.	The report records the required sequence; no stash entry remains after pop.
A6 rebase	Reflog: rebase started onto hotfix 570787f and finished with dev at 6e78009.	Dev was replayed on top of updated main.
A7 reset	Reflog records soft reset and hard reset on scratch-reset-demo; branch ends at Throwaway commit 1.	Soft retained staged changes; hard discarded commit and worktree changes.
A8 mv/rm	Commits 7e2b603 and 6e78009 retain preparation and reorganization events.	Tracked move/removal history is retained.

Representative command evidence

```
$ git log --oneline --graph --all
* 932d0d5 (HEAD -> main, origin/main) Docs: update report / Drive remote
* f3fe902 Docs: add PDF report and generator
* d76cdaf E5: pipeline reproducible from merged state
* dcc7948 E5: resolve code and DVC data conflicts
| * c7c77ec (teammate-sim) E1: normalization [-1,1]
* | 49a91eb E2: main Z-score normalization
|/
* b81c629 (tag: v2, dev) D5: dense_units=256, accuracy=0.8781
* faac07e (tag: v1) D3: dense_units=128, accuracy=0.8740

$ git reflog --all | evidence summary
6e78009 rebase (finish) onto 570787f
fa5cd83 -> d5399cf reset HEAD~1 (soft demonstration)
fa9b02b -> d5399cf reset HEAD~1 (hard demonstration)
```

Part B - Modular TensorFlow ANN Pipeline

Script	Inputs	Outputs / verified behavior
src/prepare.py	tf.keras Fashion-MNIST	Four raw .npy arrays; 60k training and 10k test examples.
src/preprocess.py	raw arrays, params.yaml	Float32 [0,1] scaling; stratified 20% validation split; six processed arrays.
src/train.py	processed arrays, train params	Flatten -> Dense(ReLU) -> Dropout -> Dense(10, Softmax); Adam; model.h5 and history.csv.
src/evaluate.py	model + processed test set	Loss/accuracy in metrics.json and labeled confusion matrix PNG.

Part D - Reproducible DVC pipeline

params.yaml

```
preprocess:
  test_size: 0.2
  seed: 42

train:
  dense_units: 256
  dropout_rate: 0.2
  learning_rate: 0.001
  epochs: 10
  batch_size: 64
```

dvc.yaml

```
stages:
  prepare:
    cmd: python3 src/prepare.py
    deps:
      - src/prepare.py
    outs:
      - data/raw

  preprocess:
    cmd: python3 src/preprocess.py
    deps:
      - data/raw
      - src/preprocess.py
    params:
      - preprocess.test_size
      - preprocess.seed
    outs:
      - data/processed

  train:
    cmd: python3 src/train.py
    deps:
      - data/processed
      - src/train.py
    params:
      - train.batch_size
      - train.dense_units
      - train.dropout_rate
      - train.epochs
      - train.learning_rate
    outs:
      - models/history.csv
      - models/model.h5

  evaluate:
    cmd: python3 src/evaluate.py
    deps:
      - data/processed
      - models/model.h5
      - src/evaluate.py
    outs:
      - reports/confusion_matrix.png
    metrics:
      - metrics.json:
          cache: false
```

D3/D4 execution and caching

At v1 (tag faac07e), a clean reproduction created dvc.lock with dense_units=128 and test accuracy 0.8740. For v2 (tag b81c629), only **train** and downstream **evaluate** were invalidated by dense_units 128 -> 256; prepare and preprocess remained cached because their dependencies and parameters did not change.

Version	Dense	Dropout	LR	Epochs	Loss	Accuracy	Target
v1	128	0.20	0.001	10	0.3411	87.40%	Met
v2	256	0.20	0.001	10	0.3474	87.81%	Met
Delta	+128	-	-	-	+0.0063	+0.41 pp	Met

Current lock-file evidence

```
prepare      raw hash 585a7ccf...dir
preprocess   processed hash 673bfbf0...dir (test_size=0.2, seed=42)
train        model hash bf4f29dc... (dense_units=256)
evaluate     metrics hash 94391a7f...; confusion matrix hash 3469b359...
metrics.json test_accuracy=0.87809998, test_loss=0.34735397
```

Part C - DVC Setup and Google Drive Remote

DVC is initialized and the default remote is configured as **gdrive_storage**. [Open the Google Drive DVC folder](#) (folder ID: **1A4889YAEgPAiFNWejJkhAuGMxRLowjNe**). A dedicated Google Cloud desktop OAuth client was authorized without committing credentials to Git. The final **dvc push** uploaded 14 files, **dvc status -c** reported the cache and remote in sync, and the supplied Drive folder visibly contains the DVC **files** directory.

Check	Status	Evidence / required action
DVC initialized	Verified	.dvc exists and dvc version identifies the repository.
Default Drive remote	Verified	.dvc/config points to the supplied Drive folder ID.
Credentials excluded	Verified	.gitignore excludes credentials.json, client_secrets.json, token.json and token patterns.
dvc push	Verified	Completed successfully: 14 files pushed.
Drive contents	Verified	Google Drive shows the DVC files directory; remote status is in sync.
Instructor sharing	User action required	Share the folder with the instructor account if it is not already shared.

Part E - Simulated Collaboration and Conflict Resolution

The conflict is genuine: teammate-sim used zero-centered [-1,1] scaling while main independently used Z-score normalization. Both branches regenerated data/processed, yielding different DVC hashes. Merge commit dcc7948 reconciled the code to canonical [0,1] scaling and selected regenerated processed hash 673bfbf0...dir; d76cdf then recorded the reproducible final state.

```
[CONFLICT DEMONSTRATION: Part E3]
$ git merge teammate-sim
Auto-merging data/processed.dvc
CONFLICT (content): Merge conflict in data/processed.dvc
Auto-merging src/preprocess.py
CONFLICT (content): Merge conflict in src/preprocess.py
Automatic merge failed; fix conflicts and then commit the result.

--- data/processed.dvc ---
<<<<<< HEAD (main)
- md5: dec45f84294f6511cbd8d5c0e9997e7f.dir
=====
+ md5: 31645b3bbc540c83585eb31cb97a35f4.dir
>>>>>> teammate-sim

--- src/preprocess.py ---
<<<<<< HEAD (main: Z-score normalization)
mean = np.mean(train_images)
std = np.std(train_images) + 1e-7
train_images = (train_images.astype('float32') - mean) / std
test_images = (test_images.astype('float32') - mean) / std
=====
+ teammate-sim: Zero-centered [-1, 1] normalization
train_images = (train_images.astype('float32') - 128.0) / 128.0
test_images = (test_images.astype('float32') - 128.0) / 128.0
>>>>>> Teammate-sim
```

Conflict evidence: Git markers are visible in both `src/preprocess.py` and `data/processed.dvc`. The final merge commit has two parents (49a91eb and c7c77ec), and the working source no longer contains conflict markers.

Final Artifacts and Compliance Audit



The confusion matrix corresponds to the v2 test accuracy of 87.81%. The weakest class separation is Shirt versus visually similar upper-body categories; this is expected for a fully connected ANN and does not affect the assignment threshold.

Submission checklist

Deliverable	Status	Notes
GitHub repository + unsquashed history	Verified	origin/main is synchronized; v1/v2 and collaboration branches exist remotely.
Four standalone pipeline scripts	Verified	All required scripts and model architecture are present.
params.yaml + dvc.yaml + dvc.lock	Verified	Four stages are wired; final lock is committed.
Accuracy >= 85%	Verified	v2 accuracy 87.81%.
Google Drive DVC objects	Verified	14 files pushed; cache and gdrive_storage are in sync.
Instructor Drive access	User action required	Share the Drive folder with the instructor account.
Report accuracy	Corrected	Repository, metrics, upload, and Drive evidence were cross-checked.

Final conclusion. The Git history, ANN implementation, DVC pipeline definition, lock file, metrics, tags, conflict-resolution history, Google Drive upload, and remote synchronization meet the technical requirements. The only remaining submission-side check is to ensure the supplied Drive folder is shared with the instructor account.